

QMC and Chiral EFT: Uncertainty Quantification and Perturbation Theory

Ryan Curry

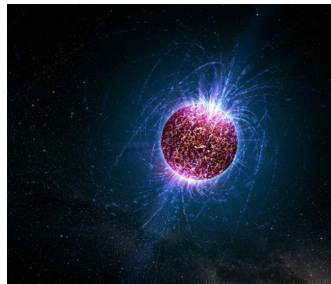
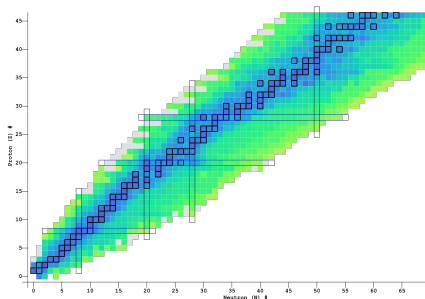
University of Guelph

2026-05-18

Nuclear Hamiltonians for Advancing Nuclear Physics and Beyond
INT, Seattle, WA

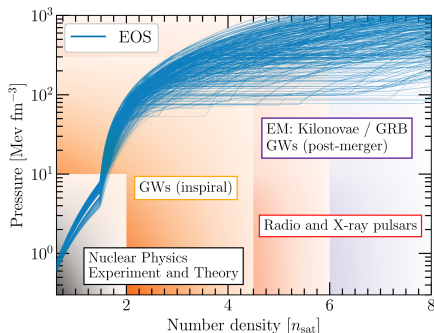
The Nuclear Many-Body Problem

From nuclear structure to neutron stars



We want to describe diverse nuclear systems from first principles

The Multi-Messenger Era



P.T.H. Pang, *et. al*
Nature Communications **14**, 8352 (2023)

- Terrestrial experiments
 - Neutron skin thickness
 - Heavy ion collisions
- Astrophysical measurements
 - Gravitational Waves
 - NICER
 - Heavy pulsars
- Nuclear Theory Calculations
- Studies analyzing these events require input from nuclear theory
- We should do everything we can to quantify and reduce our theoretical uncertainties

Solving the Nuclear Many-Body Problem

Accurate description of the
nuclear interaction

+

Powerful computational
approaches

- Effective range expansion
- Fit to experimental data
 - Argonne V18 & similar
- EFT related to underlying theory (QCD)
 - Pionless
 - Chiral (Δ -less) or -full
 - Halo
- Hartree Fock (& extensions)
- BCS Theory
- Nuclear Shell Model
- Energy Density Functionals
- No-Core Shell Model
- Quantum Monte Carlo
- Many-Body Perturbation Theory
- Coupled Cluster
- In-Medium Similarity Renormalization Group
- Self-Consistent Green's Functions

What's the plan?

QMC requires local interactions, but at $N^3\text{LO}$ this is impossible.

Can we include nonlocal operators perturbatively in our many-body calculations?

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, (2023)

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews,
Phys. Rev. C, **111**, 015801 (2025)

There are fundamental uncertainties in our description of the nuclear interaction.

What is the best way to properly account for these in our many-body calculations?

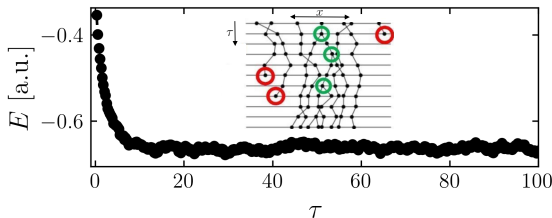
R. Curry, K. Hebeler, S. Gandolfi, A. Gezerlis,
A. Schwenk, R. Somasundaram, and I. Tews,
arXiv:2510.15860

Diffusion Monte Carlo

Solve Schrödinger equation in imaginary time $-\frac{\partial}{\partial\tau}\Psi = \left(-\frac{\hbar}{2m}\nabla^2 + V\right)\Psi$

$$\psi(\tau) = e^{-(H-E_T)\tau}\psi_T$$

- Projects out the ground-state energy



- Can straightforwardly handle high-cutoff / hard-core interactions
- Works well for central interactions $e^{-\sum_{i<j} v(\mathbf{r}_{ij})\tau}$
- Care needed for spin/isospin dependent interactions (chiral EFT)

Green's Function Monte Carlo

How do we handle a propagator with spin/isospin degrees of freedom?

$$e^{-\sum_{i<j}[v_1(\mathbf{r}_{ij})\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2+v_2(\mathbf{r}_{ij})\boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2+\dots]}\tau$$

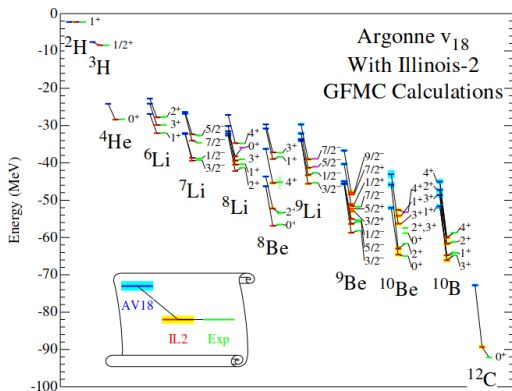
- Explicitly sum over all spin/isospin states

$$|\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ a_{\uparrow\downarrow\uparrow} \\ a_{\uparrow\downarrow\downarrow} \\ a_{\downarrow\uparrow\uparrow} \\ a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix} \quad \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 |\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ 2a_{\downarrow\uparrow\uparrow} - a_{\uparrow\downarrow\uparrow} \\ 2a_{\downarrow\uparrow\downarrow} - a_{\uparrow\downarrow\downarrow} \\ 2a_{\uparrow\downarrow\uparrow} - a_{\downarrow\uparrow\uparrow} \\ 2a_{\uparrow\downarrow\downarrow} - a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix}$$

- Each of the $2^A \frac{A!}{Z!(A-Z)!}$ spin-isospin states must be propagated
- Limiting GFMC to $A \leq 12$

Green's Function Monte Carlo

- GFMC is an exact calculation, limited to $A \leq 12$



S. Pieper, R.B. Wiringa,
Annu. Rev. Nucl. Part. Sci. **51** (2001)

- Used to study electroweak sector,
G.B. King and S. Pastore,
Annu. Rev. Nucl. Part. Sci. **74**
(2024)
- Few-body nuclear reactions
J.E. Lynn, I. Tews, J. Carlson, *et. al*, Phys. Rev. Lett., **116** (2016)
- Precision tests of standard model
G.B. King, J. Carlson, A.R. Flores,
et. al, arXiv:2509.07310

Auxiliary Field Diffusion Monte Carlo

- Instead of summing over all spin-isospin states, we can sample them
- Particles can be described by single-nucleon spinors

$$|S_i\rangle = a_i |p \uparrow\rangle + b_i |p \downarrow\rangle + c_i |n \uparrow\rangle + d_i |n \downarrow\rangle$$

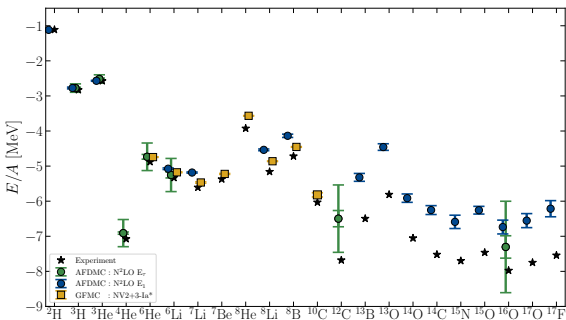
- Linearize quadratic spin/isospin ($\sigma_1 \cdot \sigma_2$, etc) dependence through Hubbard-Stratonovich transformation

$$e^{-\frac{1}{2}\lambda O^2} = \frac{1}{\sqrt{2\pi}} \int dx_e e^{-\frac{x_e^2}{2} + \sqrt{-\lambda} x_e O}$$

- We have traded two body $\sigma_1 \cdot \sigma_2$ type interactions for single particles interacting with external auxiliary fields x

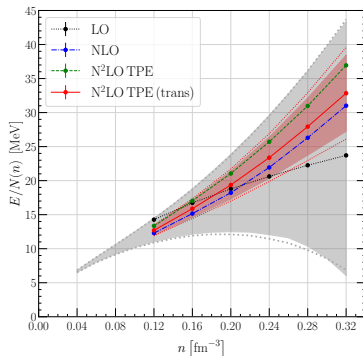
Auxiliary Field Diffusion Monte Carlo

- Trade off in precision due to sampling auxiliary fields
- Extend calculations to $A \leq 17$ as well as dense matter



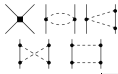
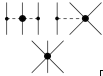
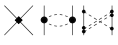
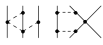
R. Curry, S. Gandolfi, A. Gezerlis, and I. Tews
PNPP, *in preparation*.

Results from D. Lonardoni, G.B. King, J. Martin



I. Tews, R. Somasundaram, *et. al*
Phys. Rev. Res., **7**, 033024 (2025)

Chiral Effective Field Theory

	2N	3N	4N
LO $\nu = 0$	  2	—	—
NLO $\nu = 2$	 7	—	—
N ² LO $\nu = 3$	 + ... 0	 2	—
N ³ LO $\nu = 4$	 + ... 12	 + ... 0	 + ... 0

- Expansion in powers of Q/Λ_b
- Separation of scales quarks vs nucleons and pions
- Power counting
- Many-body forces emerge naturally
- Reduce uncertainty by going to higher chiral order

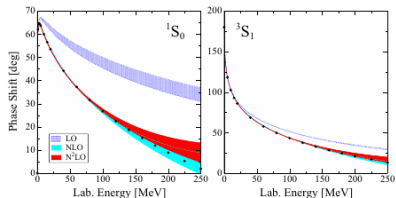
S. Weinberg, Phys. Lett. B, **251**, 288 (1990)

U. van Kolck, Phys. Rev. C, **49**, 2932 (1994)

E. Epelbaum, H.-W. Hammer, Ulf-G. Meissner,
Rev. Mod. Phys. **81**, 1773 (2009)

Local Chiral EFT

- Use only half of the operators in calculation
- Recover the rest through anti-symmetrization



- $q \rightarrow r$ local
- $k \rightarrow \nabla$ nonlocal

$$V^{(0)} = \alpha_1 + \alpha_2 \sigma_1 \cdot \sigma_2 + \alpha_3 \tau_1 \cdot \tau_2 + \alpha_4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2$$

$$\begin{aligned} V_{\text{cont}}^{(2)} = & \gamma_1 q^2 + \gamma_2 q^2 \sigma_1 \cdot \sigma_2 + \gamma_3 q^2 \tau_1 \cdot \tau_2 \\ & + \gamma_4 q^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\ & + \gamma_5 k^2 + \gamma_6 k^2 \sigma_1 \cdot \sigma_2 + \gamma_7 k^2 \tau_1 \cdot \tau_2 \\ & + \gamma_8 k^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\ & + \gamma_9 (\sigma_1 + \sigma_2)(q \times k) \\ & + \gamma_{10} (\sigma_1 + \sigma_2)(q \times k) \tau_1 \cdot \tau_2 \\ & + \gamma_{11} (\sigma_1 \cdot q)(\sigma_2 \cdot q) \\ & + \gamma_{12} (\sigma_1 \cdot q)(\sigma_2 \cdot q) \tau_1 \cdot \tau_2 \\ & + \gamma_{13} (\sigma_1 \cdot k)(\sigma_2 \cdot k) \\ & + \gamma_{14} (\sigma_1 \cdot k)(\sigma_2 \cdot k) \tau_1 \cdot \tau_2 \end{aligned}$$

Trouble on the horizon ...

- 30 new contact operators at N³LO (reduce to 12: Fierz, UT)
- Nonlocal operators cannot be totally avoided

$$\begin{aligned}
 V_{\text{cont}}^{(4)} = & \alpha_1 q^4 + \alpha_2 q^4 \tau_1 \cdot \tau_2 + \alpha_3 q^4 \sigma_1 \cdot \sigma_2 + \alpha_4 q^4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_5 k^4 + \alpha_6 k^4 \tau_1 \cdot \tau_2 + \alpha_7 k^4 \sigma_1 \cdot \sigma_2 + \alpha_8 k^4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_9 q^2 k^2 + \alpha_{10} q^2 k^2 \tau_1 \cdot \tau_2 + \alpha_{11} q^2 k^2 \sigma_1 \cdot \sigma_2 + \alpha_{12} q^2 k^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_{13} (q \times k)^2 + \alpha_{14} (q \times k)^2 \tau_1 \cdot \tau_2 + \alpha_{15} (q \times k)^2 \sigma_1 \cdot \sigma_2 + \alpha_{16} (q \times k)^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \frac{i}{2} \alpha_{17} q^2 (\sigma_1 + \sigma_2) \cdot (q \times k) + \frac{i}{2} \alpha_{18} q^2 (\sigma_1 + \sigma_2) \cdot (q \times k) \tau_1 \cdot \tau_2 \\
 & + \frac{i}{2} \alpha_{19} k^2 (\sigma_1 + \sigma_2) \cdot (q \times k) + \frac{i}{2} \alpha_{20} k^2 (\sigma_1 + \sigma_2) \cdot (q \times k) \tau_1 \cdot \tau_2 \\
 & + \alpha_{21} q^2 \sigma_1 \cdot q \sigma_2 \cdot q + \alpha_{22} q^2 \sigma_1 \cdot q \sigma_2 \cdot q \tau_1 \cdot \tau_2 + \alpha_{23} k^2 \sigma_1 \cdot q \sigma_2 \cdot q \\
 & + \alpha_{24} k^2 \sigma_1 \cdot q \sigma_2 \cdot q \tau_1 \cdot \tau_2 + \alpha_{25} q^2 \sigma_1 \cdot k \sigma_2 \cdot k + \alpha_{26} q^2 \sigma_1 \cdot k \sigma_2 \cdot k \tau_1 \cdot \tau_2 \\
 & + \alpha_{27} k^2 \sigma_1 \cdot k \sigma_2 \cdot k + \alpha_{28} k^2 \sigma_1 \cdot k \sigma_2 \cdot k \tau_1 \cdot \tau_2 \\
 & + \alpha_{29} \sigma_1 \cdot (q \times k) \sigma_2 \cdot (q \times k) + \alpha_{30} \sigma_1 \cdot (q \times k) \sigma_2 \cdot (q \times k) \tau_1 \cdot \tau_2 .
 \end{aligned}$$

- Need to develop a way to consistently treat nonlocal operators perturbatively in Quantum Monte Carlo

Tricky beyond first order

First- and Second-order Corrections

$$E_0^{(1)} = \langle \psi_0 | V' | \psi_0 \rangle$$

$$E_0^{(2)} = - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_0 | V' | \psi_k \rangle|^2}{E_k - E_0}$$

The problem?

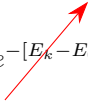
$$\lim_{\tau \rightarrow \infty} \psi(\tau) = \lim_{\tau \rightarrow \infty} e^{-(H-E_T)\tau} \psi_T \propto \psi_0 \longrightarrow E_0$$

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, L042021, (2023)

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801, (2025)

A solution - In very brief

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle$$

$$I(\mathcal{T}) = (E_0^{(1)})^2 \mathcal{T} - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_k | V' | \psi_0 \rangle|^2}{E_k - E_0} [e^{-[E_k - E_0]\mathcal{T}} - 1]$$


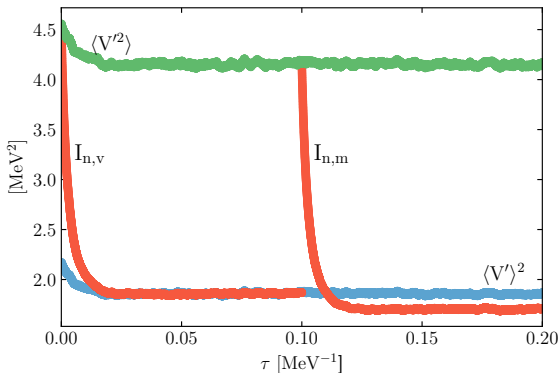
$$I(\mathcal{T} \rightarrow \infty) = (E_0^{(1)})^2 \mathcal{T} - E_0^{(2)}$$

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, L042021, (2023)

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801, (2025)

Practically speaking

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle \longrightarrow I \approx \frac{\sum_i^{\mathcal{N}} w_i V'(\mathbf{R}_i) V'(\mathbf{R}'_i)}{\sum_i^{\mathcal{N}} w_i}$$



R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, L042021, (2023)

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801, (2025)

Non-Local Operators

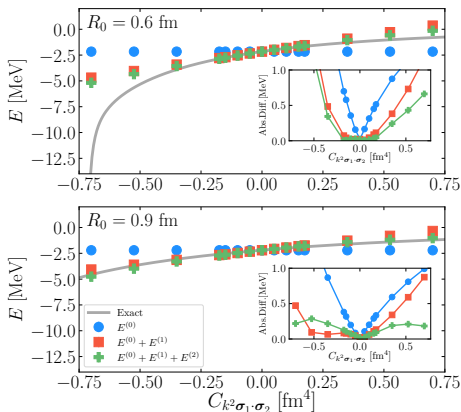
- Non-local operators cannot be avoided at higher chiral orders.
- Non-local operators cannot easily be included in imaginary time propagator
- Let's replace a local operator at NLO with a nonlocal equivalent and treat perturbatively
- $q \rightarrow r$ local
- $k \rightarrow \nabla$ nonlocal

$$V^{(0)} = \alpha_1 + \alpha_2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \alpha_3 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + \alpha_4 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2$$

$$\begin{aligned} V_{\text{cont}}^{(2)} = & \gamma_1 q^2 + \gamma_2 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_3 q^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_4 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_5 k^2 + \gamma_6 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_7 k^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_8 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_9 (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \\ & + \gamma_{10} (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{11} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \\ & + \gamma_{12} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{13} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \\ & + \gamma_{14} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \end{aligned}$$

Deuteron with nonlocal operator

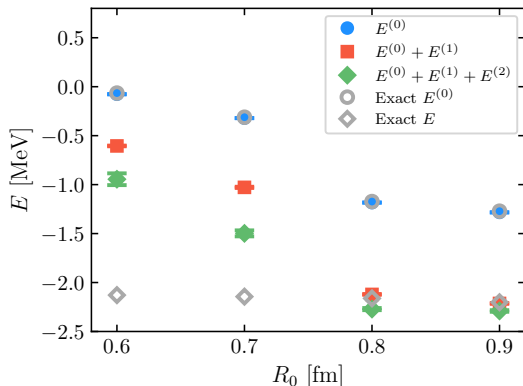
$$k^2 \longrightarrow \left[-\frac{1}{4} (\nabla^2 \delta_{R_0} \psi(\mathbf{r}) - \frac{1}{r} \frac{\partial \delta_{R_0}}{\partial r} (\mathbf{r} \cdot \nabla \psi(\mathbf{r})) - \delta_{R_0} \nabla^2 \psi(\mathbf{r}) \right]$$



- Non-local operator not included in fit to phase shifts
- $k^2 \sigma_1 \cdot \sigma_2$ included perturbatively for two cutoffs
- Compare against exact Lippmann-Schwinger solution

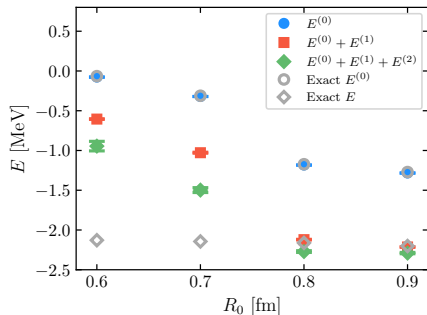
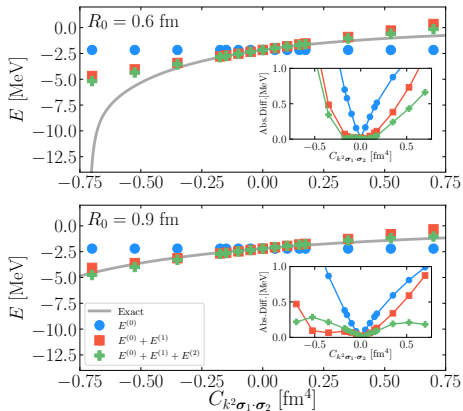
A more realistic test case

- Non-local operator included in fit to phase shifts



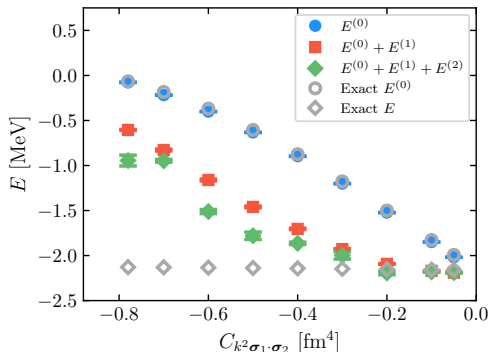
- Full range of coordinate space cutoffs
- Issues with perturbativeness at harder cutoffs

Wait a minute...



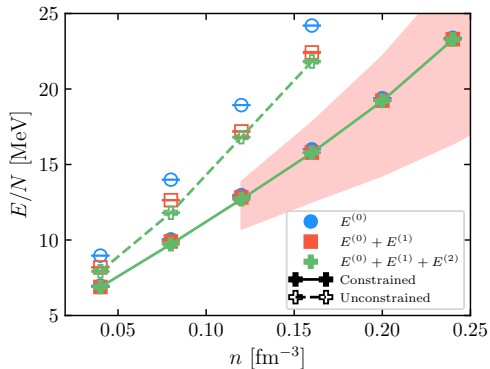
Can we take inspiration from the first case, and improve on the hard cutoff?

Constraining the nonlocal operator



- Hard-cutoff interaction
 $R_0 = 0.6$ fm ($\Lambda_c \approx 600$ MeV)
- Non-local operator included in fit to phase shifts
- Create a series of NLO interactions with a variable nonlocal LEC
- Clear region where perturbation theory is sufficient!

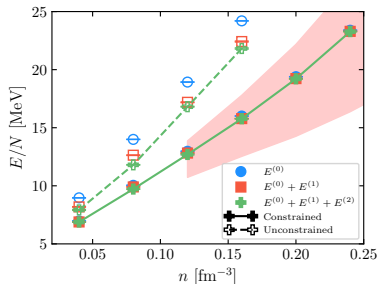
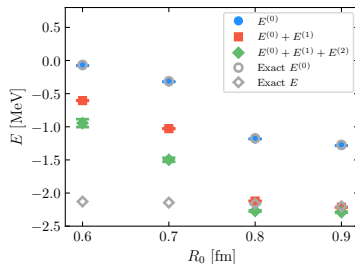
Neutron Matter



- Use constrained NLO interaction for calculation of neutron matter EOS
- Clear improvement from constraining the nonlocal operator
- Able to include nonlocal operators perturbatively even for hard cutoff interactions!

Summary

- Novel method for calculating second-order perturbation theory in a QMC context
- Treat nonlocal operators perturbatively for diverse nuclear systems
- Constrained nonlocal operator to treat perturbatively for hard-cutoff interactions
- Paves the way for perturbative inclusion of $N^3\text{LO}$ nonlocal contacts in AFDMC



There are fundamental uncertainties that should be accounted for in a many-body calculation.

- Nuclear Interaction (Experimental / Theoretical)
 - Interactions are fit to experimental measurement (which come with some associated error)
 - Chiral EFT Hamiltonian must be truncated at a finite order
- Many-Body Methods (Computational / Theoretical)
 - Model space truncation (NCSM)
 - Fixed-node / constrained-path approximation (QMC methods)
 - Retaining only two-body operators (IMSRG(2), CCSD)

Uncertainty Propagated Quantum Monte Carlo

- Train emulators on exact few-body calculations
- Bayesian inference fit of all 2N and 3N LECs at N²LO
- Propagate relevant uncertainties to many-body predictions

Quantum Monte Carlo Calculations of Light Nuclei with Fully Propagated Theoretical Uncertainties

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Achim Schwenk,^{3,4,5} Rahul Somasundaram,² and Ingo Tews²

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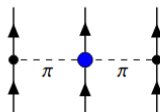
⁵*Max-Planck-Institut für Kernphysik, Saupfercheckweg 1, 69117 Heidelberg, Germany*

We report on the first quantum Monte Carlo calculations of helium isotopes with fully propagated theoretical uncertainties from the interaction to the many-body observables. To achieve this, we build emulators for solutions to the Faddeev equations for the binding energy and Gamow-Teller matrix element of ³He, as well as for auxiliary-field diffusion Monte Carlo calculations of the ⁴He charge radius, employing local two- and three-body interactions up to next-to-next-to-leading order in chiral effective field theory. We use these emulators to determine the posterior distributions for all low-energy couplings that appear in the interaction up to this order using Bayesian inference while accounting for theoretical uncertainties. We then build emulators for auxiliary-field diffusion Monte Carlo for helium isotopes and propagate the full posterior distributions to these systems. Our approach serves as a framework for *ab initio* studies of atomic nuclei with consistently treated and correlated theoretical uncertainties.

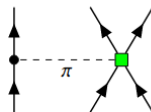
arxiv:2510.015860

Fitting chiral EFT interactions

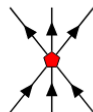
- 9 two-body LECS (C_S, C_T, C_{1-7}) up to N^2 LO fit against experimental phase shift data
- Long-range 3N couplings c_1, c_3 , and c_4 are constrained by π N scattering data.
- Short-range 3N couplings c_D and c_E need to be constrained by few-body observables.



$V_{3N}^{2\pi}(c_1, c_3, c_4)$



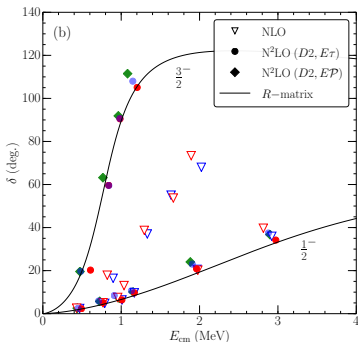
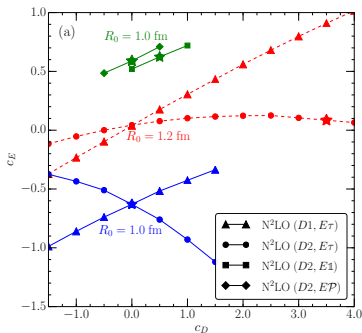
$V_{3N}^{1\pi}(c_D)$



$V_{3N}^{\text{contact}}(c_E)$

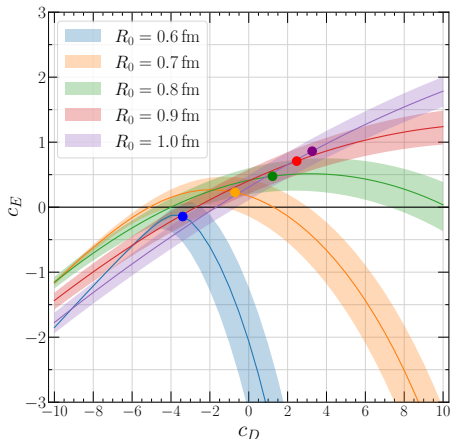
Previously for local interactions

- c_D and c_E fit to GFMC binding energy of ^4He and $n\text{-}\alpha$ p -wave phase shifts
- Using a least-squares minimization approach



More recently...

- Fit against Faddeev calculations
- ^3H binding energy and β -decay (similar to Gazit *et. al* 2009)
- These calculations don't mix well with Bayesian inference
- A single Faddeev calculation takes ~ 1 hour
- Bayesian inference requires 10^5 - 10^6 evaluations



A Tale of Two Emulators

Algorithm that mimics the behavior of a high-fidelity calculation for a fraction of the computational cost.

- In our case:
 - High fidelity means *ab initio* few-body calculation
 - Computational cost $\sim 10^3$ s (Faddeev) or 10^6 s (AFDMC)
- The goal:
 - Eigenvector Continuation (EC)
 - Parametric Matrix Model (PMM)
 - Dramatic reduction in computational cost

D. Frame, R. He, I. Ipsen, D. Lee, D. Lee, and E. Rrapaj Phys. Rev. Lett. **121**, 032501 (2018)

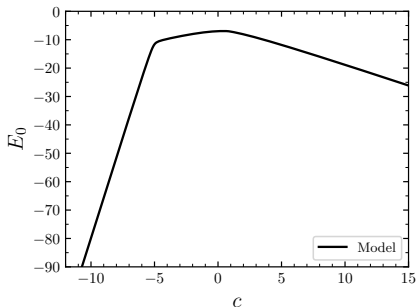
P. Cook, D. Jammooa, M. Hjorth-Jensen, D. Lee, and D. Lee, Nat. Commun. **16**, 5929 (2025)

An example...

Consider a “high-fidelity model” with a single control parameter c

$$H = H_0 + cH_1 \quad (1)$$

We can solve for the lowest eigenvalue of H for any value of c
(but let's pretend it's very expensive)

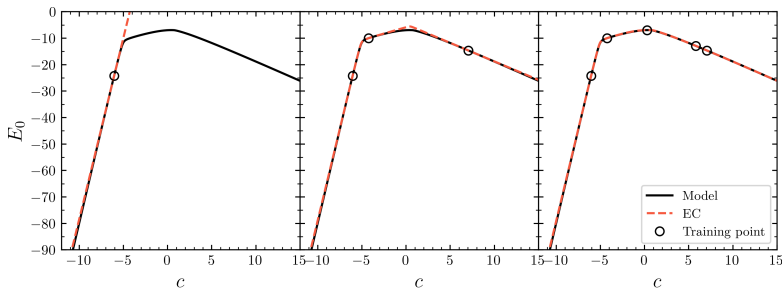


Eigenvector Continuation

- Projects the Hamiltonian to a subspace where its ground state can be rapidly evaluated
- For two training points c_1 and c_2 , with eigenvectors $|\psi_1\rangle$ and $|\psi_2\rangle$

$$M = \begin{bmatrix} \langle \psi_1 | H_0 | \psi_1 \rangle & \langle \psi_1 | H_0 | \psi_2 \rangle \\ \langle \psi_2 | H_0 | \psi_1 \rangle & \langle \psi_2 | H_0 | \psi_2 \rangle \end{bmatrix} + c \begin{bmatrix} \langle \psi_1 | H_1 | \psi_1 \rangle & \langle \psi_1 | H_1 | \psi_2 \rangle \\ \langle \psi_2 | H_1 | \psi_1 \rangle & \langle \psi_2 | H_1 | \psi_2 \rangle \end{bmatrix}$$

- Solve for smallest eigenvalue in the projected subspace for all values of the control parameter



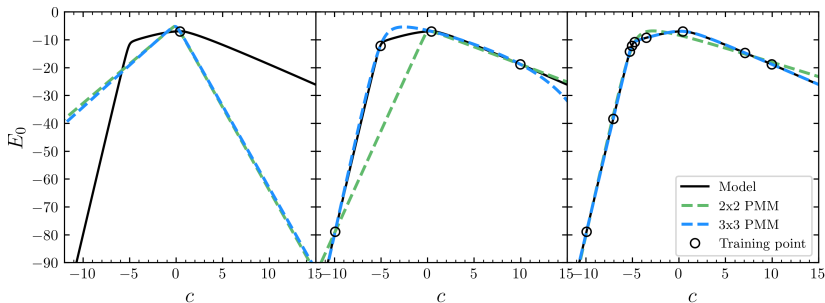
- Exact eigenstate requirement very tricky for AFDMC

Parametric Matrix Model

- Instead of explicit overlaps, fit matrix elements by minimizing cost function
- Dimensionality of PMM matrices is a free hyperparameter
- Can train against any number of training points

$$\tilde{H} = \begin{bmatrix} x_1 & 0 \\ 0 & x_2 \end{bmatrix} + c \begin{bmatrix} x_3 & x_4 \\ x_4 & x_5 \end{bmatrix}$$

- Does not require any knowledge about exact eigenstates of the high-fidelity model (i.e. can be straightforwardly used for QMC)

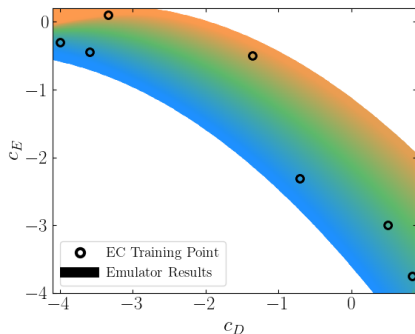
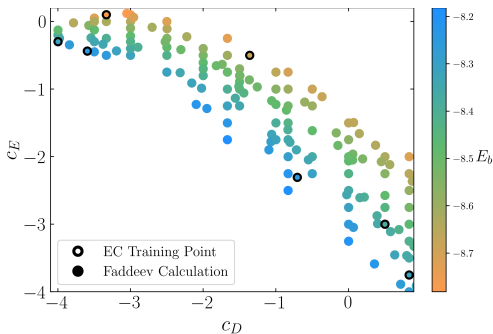


Emulating Solutions to the Faddeev Equations

$$H = T + V_{NN} + V_{3N}^{TPE} + V_{3N}^D + V_{3N}^E$$

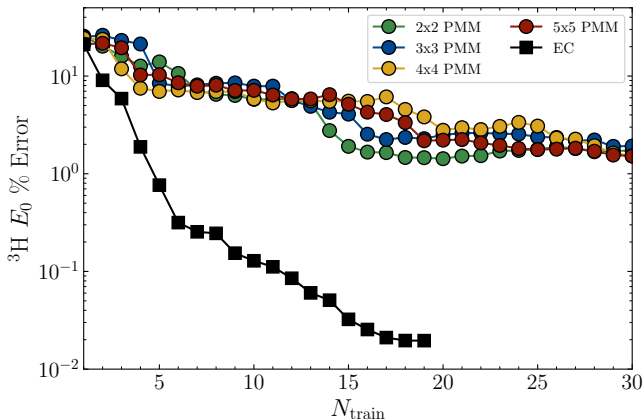
$$\tilde{H} = H_0 + C_S H_S + C_T H_T + \sum_{i=1}^7 C_i H_i + c_D H_D + c_E H_E$$

- High-cutoff interaction $R_0 = 0.6$ fm or $\Lambda_c \approx 660$ MeV
- Faddeev equations give exact solution for three-body wavefunction
- Solve for E_0 of ${}^3\text{H}$ for ~ 100 LEC sets (c_D/c_E shown for example)



Parametric Matrix Model vs Eigenvector Continuation

- Study how % error decreases with number of training points
- Additional computational overhead for eigenvector continuation (*seemingly* worth the effort)

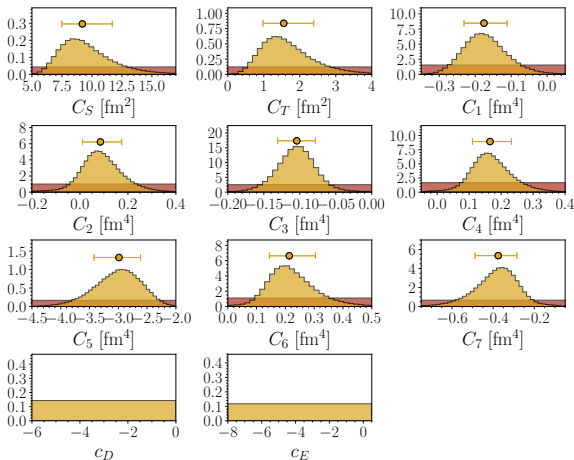


$$\mathcal{L} \propto \prod_i \exp \left[-\frac{1}{2} \left(\frac{X_i^{\text{exp.}} - X_i^{\text{theo.}}}{\sigma_i} \right)^2 \right]$$

- X_i includes ^3H binding energy, GT matrix element (Faddeev) and ^4He charge radius (AFDMC)
- $\sigma_i^2 = \sigma_{i,\text{exp}}^2 + \sigma_{i,\text{theo}}^2$
- Theoretical uncertainty $\sigma_{i,\text{theo}} = \Delta X_i$, estimated by performing calculations at lower chiral orders

$$\Delta X_{\text{N}^2\text{LO}}^{\text{EKM}} = \max \left[Q^4 |X_{\text{LO}}|, \quad Q^2 |X_{\text{LO}} - X_{\text{NLO}}|, \quad Q |X_{\text{NLO}} - X_{\text{N}^2\text{LO}}| \right]$$

Determining full N²LO LEC Posteriors

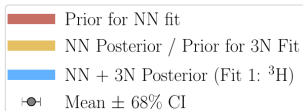
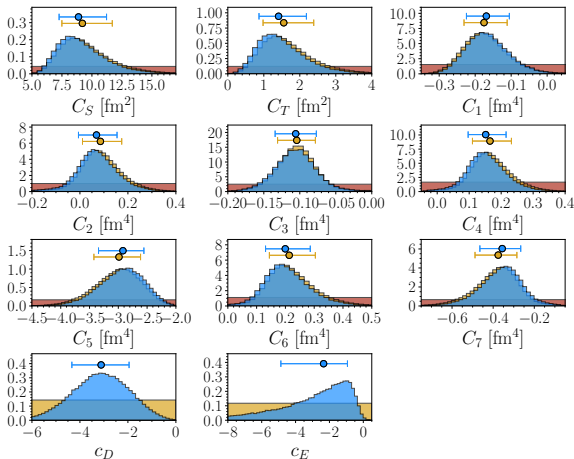


- Somasundaram *et. al* performed NN Bayesian fit at N²LO
- We take their posteriors as our priors for the NN LECs
- Include additional uniform priors on c_D and c_E

R. Somasundaram, J.E. Lynn, L. Huth, A. Schwenk, and I. Tews, Phys. Rev. C. **109**, 034005 (2024)

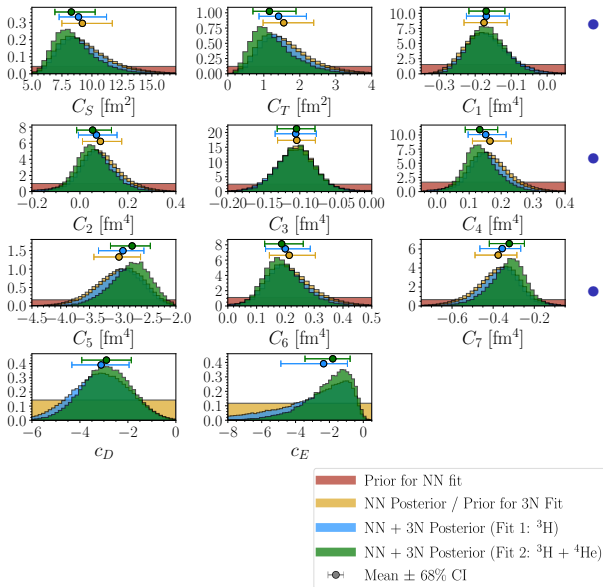
R. Curry, K. Hebeler, S. Gandolfi, A. Gezerlis, A. Schwenk, R. Somasundaram, and I. Tews
arXiv:2510.15860

Bayesian Fit Against ^3H



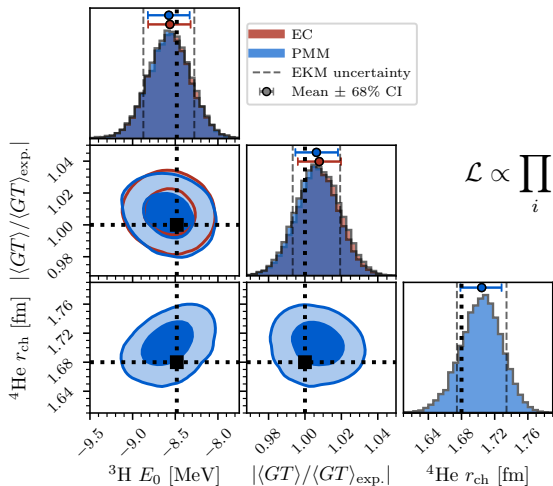
- N²LO posteriors when fit only against triton observables
- Likelihood factorizes, making this a global fit
- Very slight changes in NN LECs (strong dependence on prior)
- Long tail on c_E distribution

Additional ^4He Constraint



- AFDMC ^4He charge radius added to reduce long tail in c_E
- All NN LECs are also narrower due to its inclusion
- With LEC posteriors in hand, we can use them for many-body calculations

Posterior Predictions: PMM vs EC

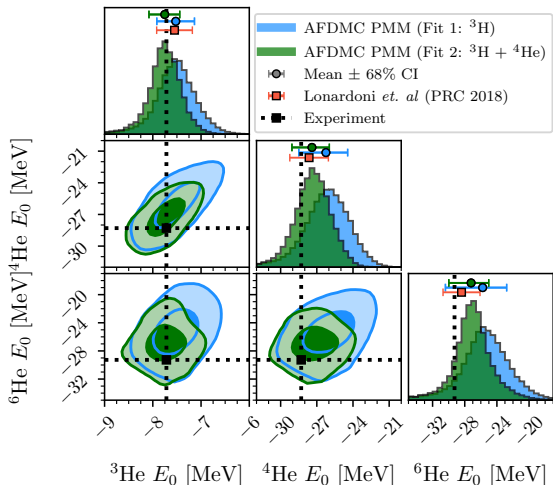


$$\mathcal{L} \propto \prod_i \exp \left[-\frac{1}{2} \left(\frac{X_i^{\text{exp.}} - X_i^{\text{theo.}}}{\sigma_i} \right)^2 \right]$$

- PMM $\equiv$ minimizing cost function
- EC $\equiv$ explicit overlap calculations

- PMM/EC give essentially identical results when used in likelihood

AFDMC calculations of helium isotopes



- Evaluate AFDMC emulators over 2.5×10^5 LEC samples
- Full posterior predictions for helium isotope binding energies
- Uncertainties propagated to many-body predictions
- Able to study correlations between different nuclei with QMC

- First Bayesian fit of local high-cutoff chiral EFT interactions containing all two- and three-nucleon operators at $N^2\text{LO}$
- Emulators employed for fitting the interaction and for the final many-body predictions
- Opportunity to easily explore other combinations of fitting observables
- Framework for future Quantum Monte Carlo studies of more light- and medium-mass nuclei with fully propagated theoretical uncertainties

Thank you for your attention!

Collaborators:

- Alex Gezerlis (Guelph)
- Ingo Tews (LANL)
- Stefano Gandolfi (LANL)
- Rahul Somasundaram (LANL)
- Achim Schwenk (TUD)
- Kai Hebeler (TUD)
- Joel Lynn (TUD)
- Kevin Schmidt (ASU)

Funding / Computational Resources:

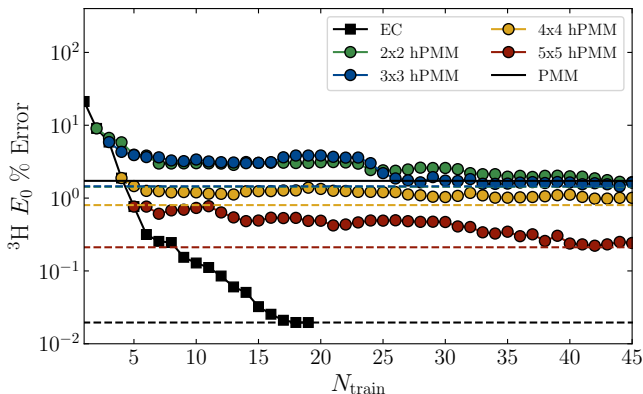


Some questions

- How can we move towards treating chiral EFT consistently (perturbative vs non-perturbative where appropriate)
- Additional few-body observables for fitting 3N interactions? Is it kosher to use medium mass nuclei, etc when fitting 3N force?
- What lessons can we learn from renormalizable theories that can be applied to chiral EFT?

Hybrid PMM-EC Emulator

- hPMM emulator starts from an EC starting matrix, and trains on additional data as a PMM
- All hPMM tests outperform previous best PMM (solid black line)
- 5×5 hPMM outperforms 8×8 EC



R. Curry, J. Kozar, C.L. Armstrong, K. Hebeler, S. Gandolfi, A. Gezerlis, A. Schwenk, R. Somasundaram, and I. Tews, *in preparation*